



SwaggerPilot

Automated OpenAPI Audit Engine

DEMO GUIDE — CONFIDENTIAL

Step-by-step walkthrough for interviews and client demos

AI Test Gen

Schema Diff

Flakiness

OWASP Security

Java Export

Vishal Mahajan • SDET 2 • 2026

01 Before You Start

Pre-demo checklist (do this 10 minutes before)

- ✓ Backend is running — terminal shows "Nest application successfully started"
- ✓ Frontend is running — <http://localhost:5173> opens in browser
- ✓ Internet connection is stable (Petstore API is external)
- ✓ Browser window is maximised — audience needs to see the full UI
- ✓ Gemini API key is in `backend/.env` as `GEMINI_API_KEY=...`
- ✓ Close Slack, Teams, email — no notification popups during demo

A Start the Backend

Open a terminal in the `swagger-pilot` folder and run:

```
cd backend && npm run start:dev
```

Expected: You see: `[Nest] LOG [NestApplication] Nest application successfully started`

! If you see "Cannot find module dotenv" — run: `npm install dotenv first`

B Start the Frontend

Open a SECOND terminal (keep backend running) and run:

```
cd frontend && npm run dev
```

Expected: Terminal shows: `VITE ready in ~500ms` → Local: <http://localhost:5173>

CLICK

Open <http://localhost:5173> in Chrome — you should see the SwaggerPilot cockpit UI

02 The Demo Flow (follow this exact order)

Total demo time: 8-10 minutes. You will cover 6 features. Each step below tells you exactly what to click and what to say.

1 Enter the API spec URL

SAY

"This is SwaggerPilot. You point it at any OpenAPI spec and it auto-generates and runs a complete test suite — no manual test writing."

CLICK

Paste this into the Swagger URL field:
<https://petstore.swagger.io/v2/swagger.json>

CLICK

Leave Base URL empty (auto-detected from spec)

CLICK

Auth Type → None (Petstore is public)

Expected: The URL field shows the Petstore URL in cyan

2 Configure the run settings

SAY

"I can control exactly what kind of audit I want. For this demo I'll turn on all the advanced features."

CLICK

Run Profile → Full (tests all categories, not just smoke)

CLICK

Chain Tests toggle → ON (tests POST then GET then DELETE flows)

CLICK

OWASP Security Probes toggle → ON (runs Top 10 security checks)

! Leave IDOR OFF — that needs two user accounts

3

Start the audit

SAY

"Watch what happens — it parses the spec, generates test cases using AI, and runs them all live."

CLICK

Click the RUN AUDIT button

Expected: Status updates stream live: Parsing spec... Generating tests... Running 80+ tests...

SAY

"It found all the endpoints automatically. No configuration, no writing test files. Just point and shoot."

Expected: After 60-90 seconds: the full report appears

03 Walking Through the Report

4 Summary Tab (show first — biggest impact)

SAY

"This is the full audit summary. Pass rate, failure count, and a release readiness verdict — go, warn, or no-go."

CLICK

Point to the RELEASE READINESS badge at the top

SAY

"It tells the team in plain English whether this API is safe to release. No spreadsheet, no manual analysis."

CLICK

Point to the Spec Coverage percentage

SAY

"It tells you exactly which percentage of your documented endpoints and status codes were actually tested."

CLICK

Point to the By Endpoint breakdown table

SAY

"Every endpoint, pass/fail count, one glance."

5 Explorer Tab — Schema Diff (the "wow" feature)

CLICK

Click the Explorer tab

CLICK

Click on any FAILED test in the list

SAY

"Look at this — it's not just telling me the test failed. It's showing me a git-diff of the response body vs the OpenAPI schema."

CLICK

Point to the red minus rows in the schema diff panel

SAY

"Red means a required field is missing from the response. Amber means the type is wrong. Green means extra undeclared fields. This tells the developer exactly what to fix."

SAY

"No other tool does this — they just say FAIL. We say WHY it failed, at the contract level."

6

Flakiness Tab

CLICK

Click the Flakiness tab

SAY

"Any test that failed was automatically re-run 5 times. These are the results. STABLE means it consistently fails — real bug. FLAKY means it sometimes passes — non-deterministic, environment issue."

SAY

"This saves hours of debugging. You immediately know which failures to escalate and which to investigate locally."

04 Security & Exports (the closer)

7 Security Tab — OWASP Top 10 (the "wtf" moment)

CLICK Click the Security tab

"This is the feature nobody else has. It ran OWASP API Security Top 10 probes — injection, mass assignment, CORS misconfiguration, missing security headers, rate limiting — all spec-aware."

SAY

CLICK Point to the Security Score gauge (0-100)

"Every finding is mapped to the OWASP category, severity level, and a specific fix recommendation."

SAY

CLICK Point to any finding card and read the Fix line

"ZAP and Burp scan generic crawls. This knows your API contract and targets the probes specifically at your endpoints and parameters. That's the difference."

SAY

8 Export Java Tests

SAY

"Here's where it closes the loop. I've discovered all these test cases. Now I'll export them as a runnable Java test suite."

CLICK Scroll to the footer — click Export Java button (green)

Expected: A .java file downloads — GeneratedApiTest.java

SAY

"This is a complete Rest-Assured test class. Drop it into the java-tests module and run it with Maven. No rewriting, no copy-paste."

SAY

"Discovery to automation in one click."

9

Export Client PDF

SAY

"For clients or management who don't read test output, I can export a professional audit report."

CLICK

Click Client PDF button in the footer

Expected: A polished PDF downloads with findings, recommendations, and release status

SAY

"This goes straight to the client. No reformatting, no PowerPoint."

Your closing line:

"SwaggerPilot takes any OpenAPI spec and in under 3 minutes gives you: a full test suite, a schema contract diff, a flakiness analysis, an OWASP security score, and a runnable Java test class — all automated, zero manual setup. No other tool combines all of this in one workflow."

05 Troubleshooting

Petstore returns connection errors	Petstore is a public demo API and sometimes goes down. Switch to: <code>https://petstore3.swagger.io/api/v3/openapi.js</code> on
AI Analysis shows "404 error"	Gemini model deprecated. Check <code>backend/src/test-generator/gemini.service.ts</code> and confirm the model is set to: <code>gemini-2.0-flash</code>
Security tab does not appear	The OWASP Security Probes toggle must be ON before clicking Run Audit. It only appears if probes were enabled at run time.
"Cannot find module dotenv" on backend start	Run: <code>cd backend && npm install dotenv</code> Then restart the backend.
Tests all show ERROR (not FAIL)	The API server is unreachable. Check the Base URL is correct and your internet connection is working.
Frontend blank / white screen	Run: <code>cd frontend && npm install</code> Then: <code>npm run dev</code>
Export Java button does nothing	Check browser pop-up blocker – allow downloads from localhost:5173
Flakiness tab missing	Flakiness only appears when at least 1 test fails. Run against a real API with known failures.

Need help mid-demo? Keep this guide open on your phone — nobody will know.